Mohak Saxena

PCB design project
5 V, push-to-sound audible and visual alarm.

Design interpretation
The supplied brief names an Emergency Panic Alarm Board and lists a push button, buzzer, red LED, resistor and 5 V connector. One sentence refers to a water-level module; this design follows the title, scenario and listed alarm components.

Operation
A regulated 5 V supply enters J1. Holding normally-open SW1 connects +5 V to ALARM_5V. The active buzzer BZ1 sounds, and D1 illuminates through R1. Releasing the button switches both outputs off. No latching or reset stage is included.

Electrical choices
R1 is 330 ohms, rated 0.25 W. With an assumed 2 V red LED forward voltage, nominal LED current is (5 - 2) / 330 = 9.1 mA and resistor dissipation is approximately 27 mW. BZ1 is specified as a 5 V internally driven indicator, not a passive transducer.

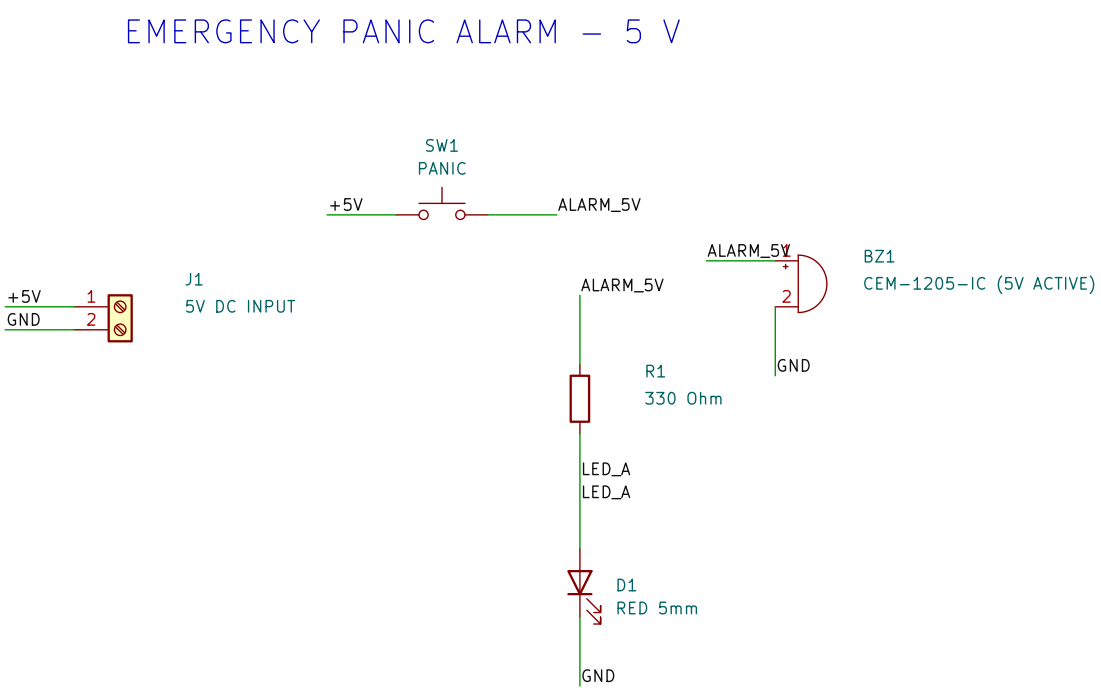
PCB
50 x 38 mm, two copper layers, through-hole assembly, 0.6 mm routed tracks and no vias. Silkscreen identifies power polarity, button function, LED color and project attribution.

Important limit
This is a design exercise, not a certified life-safety system. It has no battery backup, fault supervision or latched alarm. No physical assembly, sound-level measurement or bench testing has been performed.

Schematic

MOHAK SAXENA | EMERGENCY PANIC ALARM BOARD

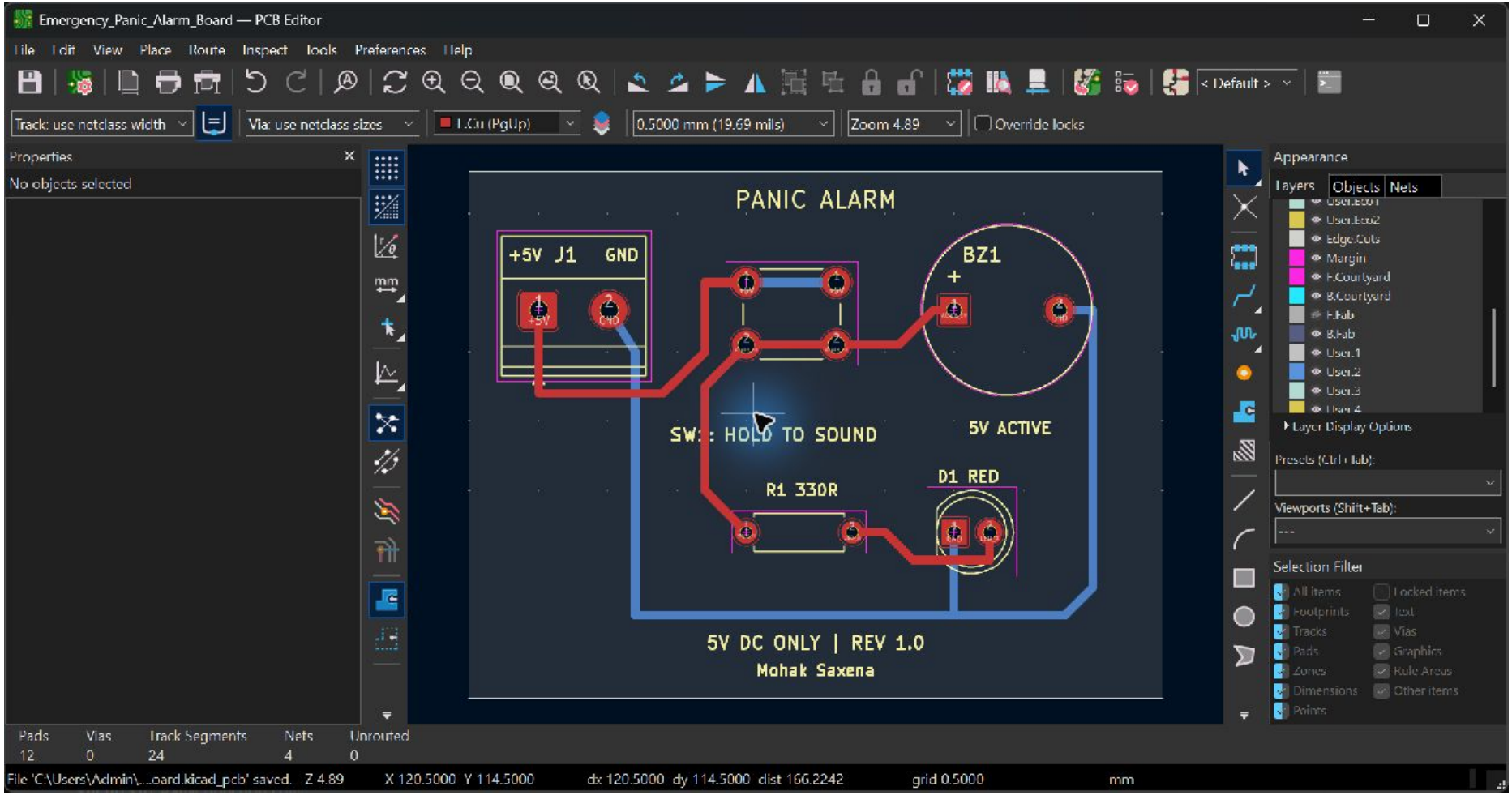
Named net labels join matching connections throughout the sheet. All five components have assigned footprints.



Footprints and Completed Routing

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KiCad PCB Editor capture: all 12 pads are connected; red traces are front copper and blue traces are back copper.

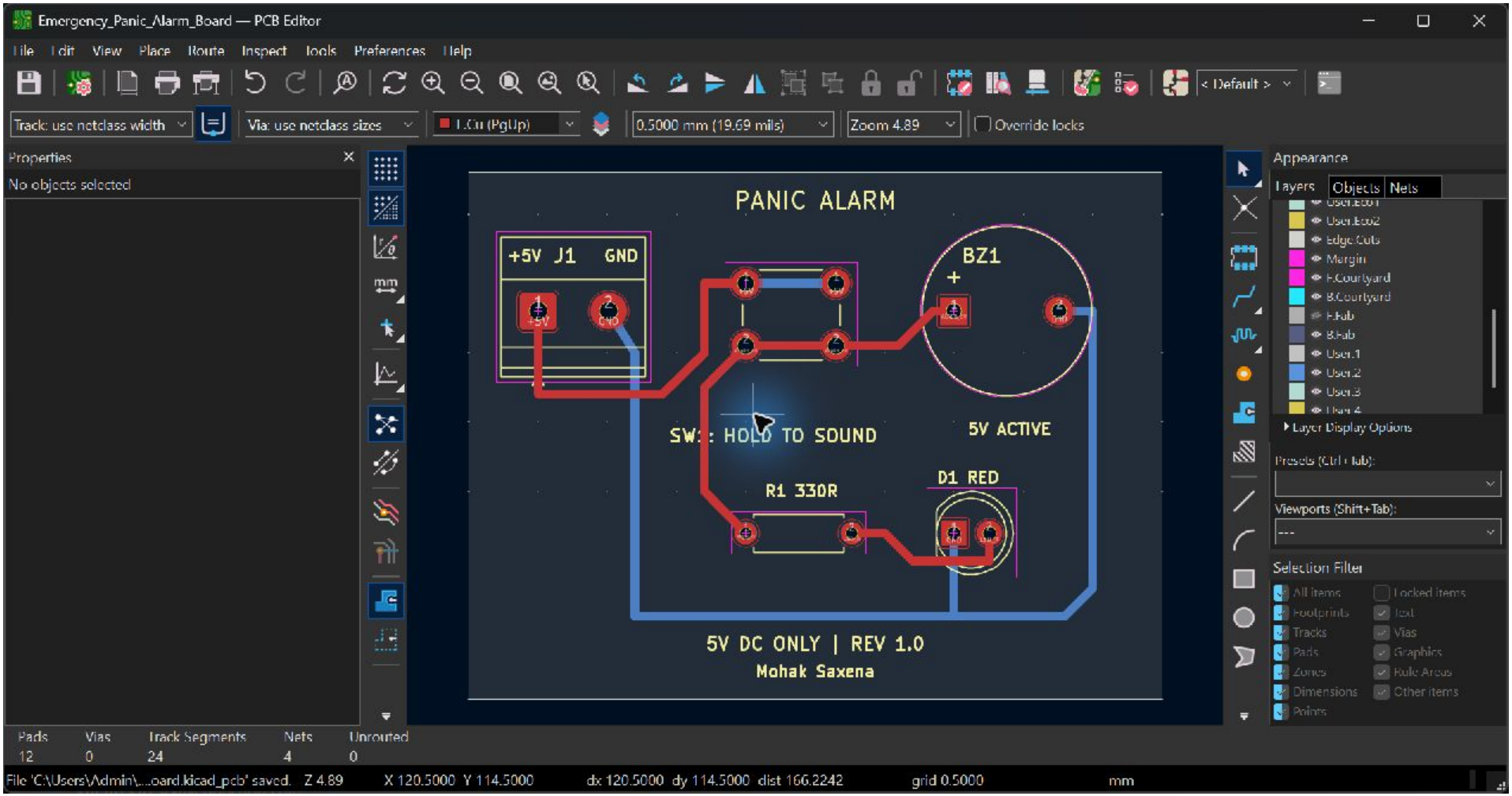


Through-hole footprints: 5.08 mm terminal block, 6 mm push button, 12 mm buzzer with 7.6 mm lead spacing, axial resistor with 7.62 mm pitch, and 5 mm LED with 2.54 mm pitch.

2D PCB Layout

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Two-layer board with a closed Edge.Cuts outline. The power connector, panic button and outputs are kept separated for access and identification.

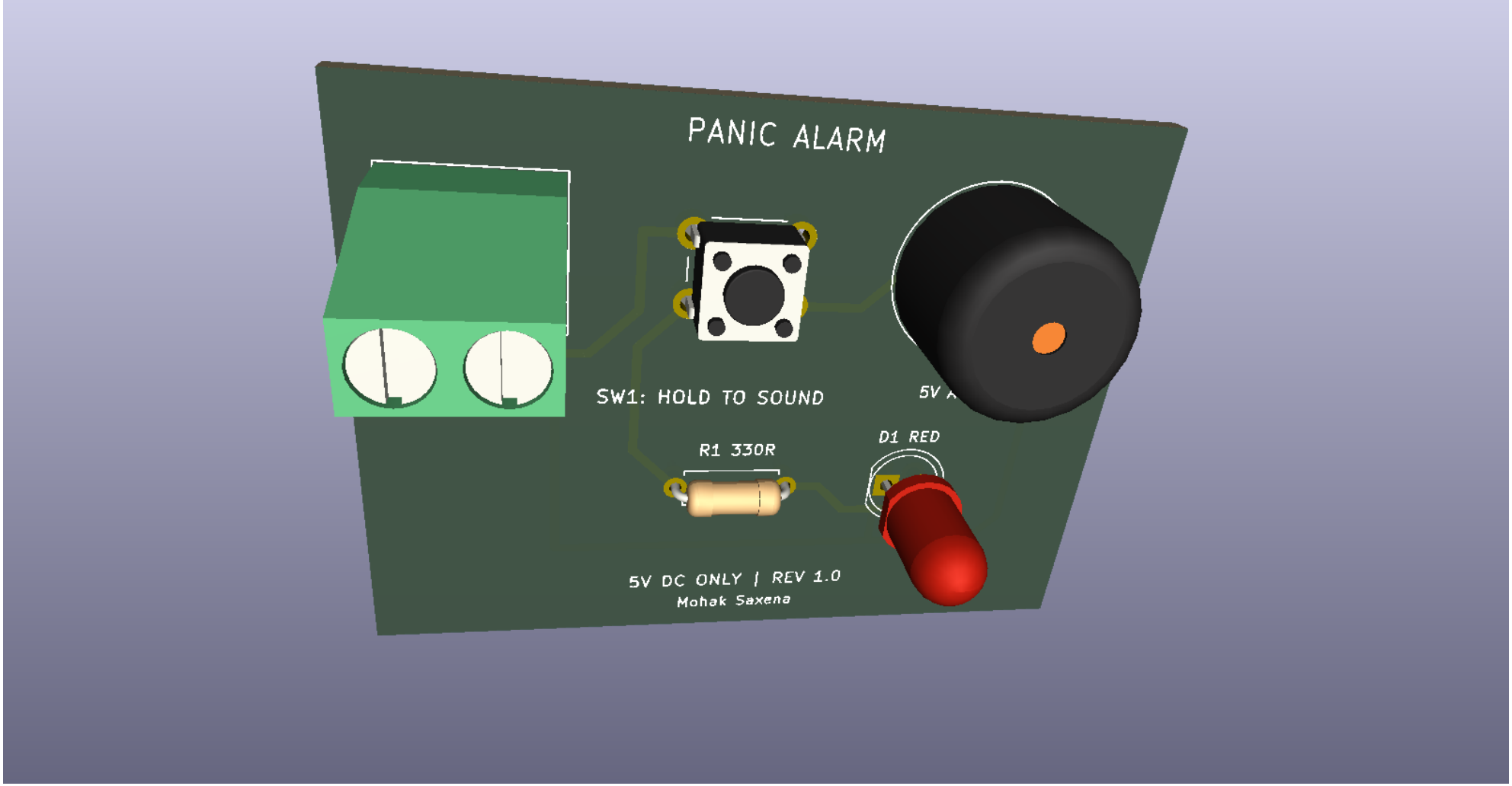


Board size: 50 x 38 mm. No mounting holes were specified. Check the intended enclosure before fabrication.

3D PCB View

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Exported from KiCad 3D Viewer after visual inspection. Stock library models illustrate assembly; they do not replace the purchased parts' mechanical drawings.



Bill of Materials and Verification

MOHAK SAXENA | EMERGENCY PANIC ALARM BOARD

Ref.	Qty	Component / required specification
J1	1	2-pin 5.08 mm screw terminal; pin 1 = +5 V, pin 2 = GND
SW1	1	Normally-open 6 mm momentary push button
BZ1	1	CEM-1205-IC, active 5 V buzzer indicator; 12 mm body, 7.6 mm pitch
R1	1	330 ohm axial resistor, 0.25 W; 7.62 mm pitch
D1	1	Red 5 mm through-hole LED, 2.54 mm pitch

KiCad checks performed

Electrical Rules Check: 0 errors and 0 warnings.
Design Rules Check: 0 errors and 0 warnings.
Unconnected items: 0. Schematic/PCB parity issues: 0.
Checks used KiCad's project/default rule settings. Passing these checks is not proof of physical performance or certification.

Fabrication outputs

Front/back copper, front/back solder mask, front/back silkscreen, Edge.Cuts and Gerber job file; Excellon plated and non-plated drill files. Empty paste layers are included because the board uses through-hole components.

Before assembly

Confirm the exact supplied part dimensions, active-buzzer polarity and voltage rating, LED polarity and switch pin grouping. Use a current-limited regulated 5 V source. Inspect for shorts before power-on; verify press/release behavior on a bench.

Reference

Same Sky, CEM-1205-IC manufacturer datasheet:
sameskydevices.com/product/resource/cem-1205-ic.pdf
Detailed references and assigned library footprints are included in Mohak_Saxena_BOM.csv.